

Seat DIP Packages into Sockets SOP

One episode seats six chips. It runs from six chips lying in a numbered anti-static tray, a board with six empty sockets, and a test chip on its rest, to six chips seated in the six sockets, each one notch to the notch mark and flat down on its socket, with the tray empty and the test chip back on its rest.

One chip is finished before the next one is started. The order for each chip never changes: **take the chip out of its pocket, turn the notch to the notch mark, set both rows of pins in the socket, press the chip home, check the seating.** No chip is pressed until both rows of pins are standing in their holes, and no chip is touched until the one before it is home and checked.

One arm does all of it. The same gripper takes every chip, turns it, sets it on its socket and presses it home. Nothing in this task has to be held, because nothing moves on its own: the tray, the board and the test chip rest all grip the table, and a chip with both rows of pins started stands on the socket on its own. Only one arm works in this cell. Do not bring a second arm into the task.

The chip is only ever held by its two bare ends, the short ends with no pins on them. The gripper never closes on a row of pins, never touches the pins from the side, and never lays a chip down on its pins anywhere but on a socket. A bent pin cannot be undone once it is bent, so every rule in this SOP is there to keep the pins straight.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one object is held at a time, no grasp is changed while an object is off the table, and no chip is ever carried over a socket it is not going into or over a chip that is already seated. Every chip travels straight forward from its pocket to its own socket.

The pins do the deciding, not the arm. A chip goes home under one straight press, or it does not go home at all. A chip that will not go down is not forced, not rocked in end to end and not pressed a second time: it is left standing where it stopped, tagged, and the episode ends. A pin that is bent is left bent. Nothing is ever straightened, prised out or seated again on camera.

Each tool goes back to its own spot before the next step starts. The counts are fixed: six chips, six sockets, one test seating, one turn, one set down and one press per chip. There is no second press, no lifting a chip back out and no swapping chips between sockets. A chip that lands off the table is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 90 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the anti-static tray, the board with all six sockets and the test socket, and the test chip rest.
3. The top of the board stays in frame the whole time, close enough that a reviewer can see the notch on a chip, the notch mark on a socket, and whether the chip body is flat down on the socket or still standing up on its pins.
4. All six tray pockets stay in frame and their numbers face the camera, so a reviewer can see which pocket each chip came out of.
5. The light falls across the board without glare on the sockets, so the two rows of holes and the pins standing in them can be seen from above.
6. The arm is at home with its gripper open.
7. The tabletop is dry and clear of anything but the items in the Materials checklist.
8. The arm reaches every spot on the table, all six tray pockets, all six sockets, the test socket and the test chip rest, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
9. The arm comes down straight from above onto every tray pocket, onto every socket and onto the test chip rest, so a chip is lifted and lowered without being tilted or swung in.
10. The arm can turn a chip a quarter turn either way over the board without the gripper touching the board, a seated chip or the tray.
11. The paths from each pocket forward to its own socket are unobstructed. Carry one chip along each path by hand during setup: it must clear the tray edge, the test chip rest and every socket it passes.

Materials checklist

1. One anti-static tray lies flat at the back of the table, straddling the midline, with six pockets in one line, numbered 1 to 6 left to right. The numbers face the camera.
2. The tray grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or put it in a holder. Nothing pins the tray during the episode.
3. Six chips, and no more, one lying in each pocket. Every chip is the same part in the same package, so any chip fits any socket and the notch is the only thing that tells one end from the other.
4. Each chip lies flat in its pocket, pins down, notch end toward the front of the table, so no chip ever has to be turned more than a quarter turn to face its socket.

5. Each pocket is wide enough for a chip to lift straight up out of it without a pin rubbing the pocket wall, and shallow enough that both bare ends of the chip stand clear for the gripper to close on.
6. Every chip starts with both rows of pins straight and standing in line, no pin splayed out, folded under or missing. Hold a chip up during setup and sight down each row: replace a chip with a pin out of line.
7. Every chip carries a notch at one end of its body, deep enough to see on camera from above.
8. The pins of every chip stand at the same spread as the socket holes. Set one chip on a socket by hand during setup: both rows must drop into their holes with no squeezing of the rows. If a row has to be squeezed in, the chips are the wrong spread and the batch is replaced.
9. One board lies flat at the middle of the table, straddling the midline, carrying six empty sockets in one line, numbered 1 to 6 left to right with the numbers facing the camera.
10. The board grips the table in its holder. Push it by hand during setup: if it slides, turns or lifts, weight it or clamp the holder. Nothing pins the board during the episode.
11. Every socket is empty at the start of the episode, and its two rows of holes are open, clean and clear of anything left in them.
12. Every socket carries a notch mark at one end, printed or moulded, all six marks facing the front of the table, so every chip goes in the same way round.
13. Each socket holds a chip under a straight press from above and lets it go under a straight lift. Seat one chip by hand during setup and lift it out again: a socket that will not take a chip, or that grips so hard the chip cannot be lifted straight out, is replaced.
14. A clear gap sits between one socket and the next, wide enough for the gripper to come down on a chip's two ends without touching the chip in the socket beside it.
15. One test socket sits at the left end of the board, marked TEST, clear of the six numbered sockets. It is empty at the start of the episode and empty again at the end.
16. One test chip lies on its own rest at the front left, clear of the board. It is marked so it cannot be mistaken for one of the six, it carries known straight pins, and it is never seated in a numbered socket.
17. The tabletop, the tray, the board holder and the chip rest are all anti-static, and the arm is bonded to the same mat. A cell that is not bonded is not usable for this task.

Workspace layout

Three rows: the chips at the back, the board in the middle, and the test chip at the front on the left. Every chip travels straight forward out of its pocket onto the socket with the same number.

Back row:

1. Tray spot, back centre on the midline, the anti-static tray lying flat with its six pockets in one line numbered 1 to 6 left to right, each chip lying pins down with its notch end toward the front.

Middle row, left to right:

2. Test socket, at the left end of the board, marked TEST and clear of the numbered sockets.
3. Board spot, centre, straddling the midline, the board lying flat in its holder with six empty sockets in one line numbered 1 to 6 left to right, every notch mark facing the front of the table.

Front row:

4. Test chip rest, front left, clear of the board's left edge and of the test socket behind it.
5. Front strip, the bare table between the board's near edge and the front table edge. It stays bare from the start of the episode to the end. No chip is ever set down on it.

A clear gap of at least a gripper's width sits between the tray and the board, and between each socket and the next, so a chip lifts straight up out of its pocket and comes down onto its own socket without the gripper touching the tray, the board or a chip already seated. No carry passes over a seated chip, over the test socket, or over a socket the chip is not going into.

Vocabulary

- **Arm assignment:** one gripper does everything: every chip, every turn, every press. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the spot the test chip returns to and lies flat on before the next step starts. The test chip is never left in the test socket, on the board, in a tray pocket or in the gripper between steps.
- **Release point:** the gripper opens over a socket when a chip is standing on it with both rows started, or over the test chip rest. Never over the tray, never over a seated chip and never over open tabletop.
- **Stable placement:** the object stands still after release, square on its surface, and does not rock, tip, slide or lean on anything.
- **Chip:** one of the six DIP packages that are seated. It is a flat body with a row of pins down each long side and a notch at one end.
- **Bare ends:** the two short ends of the chip body, the ends with no pins on them. They are the grasp: the gripper closes on these and on nothing else.
- **Pin row:** one of the two lines of pins running down a long side of the chip. A chip has two, and they go into the socket together.
- **Straight pin:** a pin standing in line with the rest of its row, pointing straight down. Every pin on a chip is straight before the chip is picked up.

- **Bent pin:** a pin pushed out sideways, folded under the body, splayed away from its row, or standing outside its row of holes when the chip is set down.
- **Notch:** the mark cut into one end of the chip body that says which way round it goes.
- **Notch mark:** the matching mark at one end of a socket. The chip's notch and the socket's notch mark are at the same end when the chip is the right way round.
- **Socket:** one of the six numbered holders on the board that a chip is seated in. It is never lifted, never turned and never touched by the gripper.
- **Rows started:** the tips of both pin rows standing in their own rows of holes, the chip sitting level up on its pins and taking its own weight, not yet pressed.
- **Chip home:** the chip body pressed down flat on the socket, level, with no gap under it anywhere round its edge and no pin still showing between the body and the socket.
- **Test chip:** the marked chip with known straight pins, seated once in the test socket at the start to prove the sockets and the press. It is never seated in a numbered socket.
- **Quarter turn:** the most any object is tilted or turned in the gripper. Taking a chip, turning it to the notch mark and carrying it all stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Check the seating with the test chip

Goal: the test chip seated once in the test socket, seen to go home flat, and back on its rest with the test socket empty.

The check happens once, before the first chip. If the press is set too shallow or too hard, every chip in the episode is bent or left standing, so the press is proved on the test chip first.

1.1 Take the test chip

- **Check:** the test socket is empty, with nothing standing in it.
- With the **gripper**, close on the two bare ends of the test chip square, clear of both pin rows, and lift it straight up off its rest until it is clear of the table.
- Keep it level, pins down, the whole way.

Check: the test chip hangs level in the gripper and does not slip, and no pin is touched by the gripper. If it hangs tilted, lay it back on its rest and take it again. Make no more than two corrective takes.

1.2 Seat it in the test socket

- Carry it over the test socket, turn it so the notch is at the same end as the socket's notch mark, and line both pin rows up over their rows of holes.
- Lower it straight down until the tips of both rows stand in their holes and the chip takes its own weight, then open the gripper and lift clear.

- Close the gripper flat and press straight down once on the middle of the chip body, until the body sits down flat on the socket, then lift straight off.

Check: the test chip is home, flat and level on the socket with no gap under it, and no pin is bent or standing outside the socket. If it does not go home under one press, or a pin bends, end the episode and correct the socket or the press during reset.

1.3 Park the test chip

- With the **gripper**, close on the test chip's two bare ends square, lift it straight up out of the test socket without tilting it, carry it back to its rest, lower it until it lies flat, release, and lift clear.

Check: the test chip lies flat on its rest, pins down, the test socket is empty, and the board has not moved in its holder. One seating check, and one only. It is not run again between chips.

Step 2: Take one chip from the tray

Goal: one chip hanging level in the gripper by its two bare ends, pins down and straight, clear of the tray.

Chips are worked in pocket order, pocket 1 first and pocket 6 last, and each one goes into the socket with the same number. The tray grips the table, so nothing holds it while the chip comes out.

2.1 Close on the two bare ends

- With the **gripper**, come down straight from above on the chip in the next pocket in order, and close on its two bare short ends, square to the body.
- The gripper closes on the ends and on nothing else. It never closes on a pin row, never comes in from the side onto the pins, and never pinches the body across its pins.

Check: the gripper is square on both bare ends and clear of both pin rows. If it has landed on a pin row, open, lift clear and come down again. Make no more than two corrective takes.

2.2 Lift it straight out

- Lift the chip straight up out of the pocket until both pin rows are clear of the tray, before any sideways move.
- Keep it level, pins down. Do not tilt it, do not drag it up the pocket wall, and do not slide it across the tray.

Check: the chip hangs level in the gripper, pins down, and does not slip, both pin rows are still straight and in line, and the tray has not moved. If a pin has been bent coming out, end the episode and let review flag it. The chip is not put back and it is not straightened.

Step 3: Line the chip up on its socket

Goal: the chip standing on its own on the socket with the notch at the notch mark and both pin rows started in their holes, not yet pressed.

This is the step the pins are saved or bent in. The chip is lined up in the air first and only then lowered, so no pin has to be pushed sideways to find its hole.

3.1 Turn the notch to the notch mark

- Carry the chip forward to the socket with the same number as its pocket: pocket 1 to socket 1, pocket 6 to socket 6.
- Hold it level above the socket, a chip's height clear of it, and turn it no more than a quarter turn until its notch end is at the same end as the socket's notch mark.
- Both notch and mark face the front of the table when the chip is right. Turn the chip in the air, clear of the board, and never on the socket.

Check: the chip's notch and the socket's notch mark are at the same end, and the chip hangs level and square over the socket. If the chip has come out of the tray the wrong way round, turn it the quarter turn that puts the notch right. If it would take more than a quarter turn, lay it back in its pocket, take it again, and turn it once. Make no more than one correction.

3.2 Set both pin rows in the socket

- Line both pin rows up over their own rows of holes, so every pin in each row stands over a hole, and hold the chip level.
- Lower it straight down until the tips of both rows stand in their holes and the chip sits up on its pins under its own weight. Stop there.
- Both rows go in together. Do not put one row in first and lean the other one in after it, and do not slide the chip along the socket to find the holes.
- Open the gripper and lift straight clear of the chip.

Check: the chip stands level on the socket on its own, notch to the notch mark, with both rows in their holes and no pin standing outside a row, splayed out or folded under the body. If a pin is outside its row, end the episode and let review flag it. The chip is not lifted off and set down again, and no pin is straightened.

Step 4: Press the chip home

Goal: the chip body flat down on the socket, level, with no gap under it and no pin bent.

One press, straight down, on the middle of the body. The chip is already standing on its own pins, so the press only has to push it down, not hold it up or line it up.

4.1 Press it straight down once

- Close the gripper flat and bring it down square on the middle of the chip body, across both rows at once.
- Press straight down in one steady move until the body sits down flat on the socket, then stop and lift straight off.
- Press straight down and nothing else. Do not press one end and then the other, do not rock the chip in, do not slide it while it is down, and do not press a second time.
- If the chip does not go home under one press, stop and leave it standing where it is. Do not press harder and do not press again.

Check: the chip is home. If it is still standing up on its pins after the press, end the episode and let review flag it. A chip that stopped short is left in place: it is not pressed again and it is not lifted out.

4.2 Check the seating

- Look along the chip from the side: the body lies flat and level on the socket with no gap under it anywhere round its edge, and no pin still showing between the body and the socket.
- Look at the chip from above: the notch is at the socket's notch mark, and no pin is bent out, splayed or folded under the body.
- Look at the socket beside it: nothing has been knocked, lifted or moved.

Check: the chip is home, the right way round, with every pin in the socket. A chip that is seated is never lifted back out on camera, whatever the check finds. If a check fails, end the episode and correct it during reset.

Step 5: Do the next chip

- Come back empty to the tray and take the chip in the next pocket in order.
- Run Steps 2 to 4 again for that chip.
- Six chips, worked one at a time from pocket 1 to pocket 6. A chip is home and checked before the next chip is touched.
- The pockets stay in step with the sockets: the chip from pocket 1 goes in socket 1 and the chip from pocket 6 goes in socket 6.
- The test chip is not seated again. The press is proved once per episode.

Check: after each chip, the tray has one fewer chip in it, one more socket is filled, and every chip already seated is still home and untouched. No chip is seated twice or skipped.

Step 6: End the episode

- Confirm the tray: all six pockets empty.
- Confirm the board: six chips, one home in each numbered socket, each one flat and level on its socket with no gap under it.

- Confirm the notches: every chip's notch at its socket's notch mark, all six the same way round.
- Confirm the pins: no pin bent out, splayed or folded under a body, and no pin standing outside a socket.
- Confirm the test socket: empty, with the test chip flat on its rest.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the six chips straight up out of their sockets by hand, one at a time, keeping each one level so no pin is prised sideways.
2. Sight down both pin rows of every chip. Set aside any chip with a bent, splayed or folded pin: it is not straightened and it is not used again.
3. Lay six chips back in the tray, one to a pocket, pins down and notch end toward the front, and replace any chip that was set aside so the tray holds exactly six.
4. Pick up any chip that fell off the table and bin it. A chip that has been on the floor is not seated again.
5. Look into all six sockets and the test socket: every row of holes open, clean and clear of a broken pin or anything else left in it. Replace a socket that holds anything.
6. Seat one chip in each socket by hand and lift it out again. Replace a socket that will not take a chip under a straight press, or that grips so hard the chip cannot be lifted straight out.
7. Confirm every socket's notch mark is still readable and all six still face the front of the table.
8. Seat the test chip in the test socket by hand, confirm it goes home flat under one straight press, lift it out and lay it flat back on its rest.
9. Confirm the tray, the board holder and the test chip rest all still grip the table, and that the tray's pocket numbers and the board's socket numbers still face the camera.
10. Wipe the tabletop, the tray and the board clean and dry, and confirm the mat and the arm are still bonded.
11. Confirm the front strip is bare and the light still falls across the board without glare on the sockets.
12. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a chip is seated before the test chip check, a chip is lowered onto a socket before its notch is turned to the notch mark, a chip is pressed before both pin rows are standing in their holes, a new chip is taken out of the tray while the last one is still standing up on its pins, or a step is skipped.
- **SOP rule broken:** Steps 1 to 5, prove the press once, then for each chip take it, turn it, set both rows, press it, check it, in that order.
- **Coaching note:** test chip first, then one chip all the way home before the next one is touched. Turn it, set it, then press it.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the tray, the board or a chip, or presses a chip. Any object taken, carried or pressed by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 6, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.
- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper presses on the tray, the board or its holder to keep it still while working, or keeps hold of a chip's ends while pressing it down, instead of letting the table and the socket do it.
- **SOP rule broken:** Steps 1 to 4, nothing in this task is held: every fixture grips the table and a chip with both rows started stands on the socket on its own.
- **Coaching note:** if something has to be held down, it was not set up right. Let go and let the fixture do its job.

Violation: Chip picked up wrong

- **Visible cue:** the gripper closes on a pin row, comes in from the side onto the pins, pinches the body across its pins, takes two chips at once, takes a chip out of pocket order, drags a chip up the pocket wall or slides it across the tray, or lifts a chip sideways before both rows are clear of the pocket.
- **SOP rule broken:** Step 2, close square on the two bare ends only, and lift the chip straight up out of its pocket before any sideways move, one chip at a time and in pocket order.
- **Coaching note:** the two bare ends and nothing else, then straight up. Everything that touches a pin costs a chip.

Violation: Chip put in the wrong way round

- **Visible cue:** a chip is lowered onto a socket with its notch at the wrong end, a chip ends up seated the wrong way round, or the chip is turned while it is standing on the socket instead of in the air above it.
- **SOP rule broken:** Step 3.1, turn the chip in the air, no more than a quarter turn, until its notch is at the same end as the socket's notch mark, and only then lower it.
- **Coaching note:** notch to the mark before anything touches the socket. All six go in the same way round.

Violation: Pin rows set down wrong

- **Visible cue:** one pin row is put in and the other leaned in after it, the chip is slid along the socket to find the holes, a pin ends up standing outside its row of holes, splayed out or folded under the body, or the chip is lifted off and set down again to find the holes.
- **SOP rule broken:** Step 3.2, line both rows up over their own rows of holes in the air and lower the chip straight down until both rows start together.
- **Coaching note:** line it up before it touches. A pin that has to be pushed sideways to find its hole is already bent.

Violation: Chip pressed wrong, or pressed more than once

- **Visible cue:** the chip is pressed before both rows are started, pressed on one end and then the other, rocked in end to end, slid while it is down, pressed harder after it stops, or pressed a second time.
- **SOP rule broken:** Step 4.1, one straight press, square on the middle of the body, across both rows at once.
- **Coaching note:** one press, straight down, in the middle. A chip that will not go home under one press is telling you something, so stop.

Violation: A seated chip worked on again

- **Visible cue:** a chip that is home is lifted, prised or levered back out of its socket, moved to another socket, pressed again, or has a bent pin straightened, pulled out or nipped off on camera.
- **SOP rule broken:** Steps 4.1 and 4.2, a seated chip is never lifted back out on camera and a bent pin is left bent.
- **Coaching note:** once it is in, it stays in. Rework belongs in the reset, not in the episode.

Violation: Chip put in the wrong socket

- **Visible cue:** a chip goes into a socket whose number is not its pocket number, two chips end up in one socket, a chip is seated in the test socket, or a chip is dropped onto a socket from height.
- **SOP rule broken:** Step 3.1, each chip goes onto the socket numbered the same as the pocket it came out of, lowered straight down from just above it.
- **Coaching note:** pocket number and socket number match, every time. The test socket is not one of the six.

Violation: A chip carried over the wrong thing

- **Visible cue:** a chip passes over a chip that is already seated, over a socket it is not going into, over the test socket, or back over the tray after it has been lifted out.
- **SOP rule broken:** Steps 2 and 3, each chip travels straight forward from its pocket to its own socket and over nothing else.
- **Coaching note:** route around, not across. A chip dropped on a seated chip bends two sets of pins.

Violation: Chip laid down off a socket

- **Visible cue:** a chip is set down on the board between the sockets, on the front strip, on the tabletop or on the tray rim, or is rested on its pins anywhere but in a pocket or on a socket.
- **SOP rule broken:** Steps 2 to 4, a chip lies on its pins only in its tray pocket or on its socket, and nowhere else.
- **Coaching note:** the pins carry the chip's weight only where they are meant to. Bare table bends them.

Violation: Bad pick, carry, or release

- **Visible cue:** an object is picked up before the last one is set down, two objects are held at once, the gripper shifts or seats its hold again while the chip is off the table, a chip slides along the table or the board with no daylight under it, a chip hangs tilted because the grip was taken off square, or an object is turned past a quarter turn or turned over in the air.

- **SOP rule broken:** Steps 1 to 4, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square on its bare ends, turned no more than a quarter turn.
- **Coaching note:** set it down to change the grip, and lift before the carry. A chip that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a chip, the tray, the board or the test chip is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 4, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the gripper opens while the test chip is still above its rest, the test chip is left in the test socket, on the board or on the tabletop, or it is still in the gripper when the next step begins.
- **SOP rule broken:** Step 1.3, the test chip returns to its rest and lies flat there before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower it until it lies flat, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried chip strikes the tray, the board, a socket or a chip already seated and knocks, tips, slides or displaces it; a chip falls off the board or out of the gripper; or a chip lands short of its socket, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 4, keep each chip on a clear travel path and release only after both rows are started and the chip is standing on its own.
- **Coaching note:** watch the whole path, not just the target. The seated chips are the thing worth steering widest of.

Violation: Manipulation outside the camera frame

- **Visible cue:** a chip is taken, turned, set down or pressed partly or wholly outside the environment camera frame, or the arm stands between the camera and the board while the notch, the pins or the seating is what has to be seen.
- **SOP rule broken:** Steps 1 to 4, every manipulation happens inside the camera frame, and the board and the chip being worked stay visible.

- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the board.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a chip still in the tray, a socket still empty, a chip still standing up on its pins, a chip in the wrong socket or the wrong way round, a bent pin left unflagged, the test chip off its rest or still in the test socket, the arm short of home, or the gripper not fully open.
- **SOP rule broken:** Step 6, confirm the tray, the board, the notches, the pins and the test chip, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective chip at episode start: one that already carries a bent, splayed or missing pin, one whose notch cannot be seen, or one whose pin rows stand at the wrong spread for the socket.
- Defective socket: one that will not take a chip under a straight press, one that grips so hard a chip cannot be lifted straight out, one with a hole blocked or a broken pin left in it, or one whose notch mark is missing or unreadable.
- Defective board or holder: one that slides, turns or lifts under a correct press, or a board that will not lie flat.
- Defective tray: one that slides, turns or lifts under a correct pick, a pocket so tight that a chip cannot lift straight out, or a pocket that holds a chip below the reach of the gripper.
- Defective test chip: one whose pins are no longer straight, or that will not go home in the test socket under a correct press.
- Static event: a mat, tray or arm that has come unbonded, or a chip handled outside the anti-static setup. Discard the chip as well as the episode.
- Glare, shadow or a light flickering over the board that stops the notch, the pins or the seating being seen. Fix the lighting before the next episode.

- Tray holding fewer or more than six chips at episode start, a socket already holding a chip, or a chip lying in its pocket with the notch turned the wrong way. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take the test chip from its rest
2. Seat the test chip in the test socket
3. Lift the test chip out of the test socket
4. Return the test chip to its rest
5. Take one chip from its tray pocket
6. Turn one chip so its notch faces the notch mark
7. Set both pin rows in one socket
8. Press one chip home
9. Check one chip is seated
10. Return the arm home and end the episode